Financial Engineering Project

# Making Concentrated Liquidity Pay

*Risk-Sensitive Reinforcement Learning on Uniswap v3*

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**Abstract**

Providing liquidity on Uniswap v3 looks passive, but it is not. A liquidity provider (LP) chooses a price range, earns fees while the price stays inside it, and loses when informed flow moves the price against it. On many large pools those losses exceed the fees earned. The LP must decide when to recenter, how wide to keep the band and how much to hedge; each choice costs gas, fees or slippage.

We study those decisions as a risk-sensitive reinforcement learning problem on three years of recorded on-chain trades from the 0.05 % (500 pips) ETH/USDC pool, with 2024 held out and frictions charged. The study asks where fees fall short of losses, why a standard learner fails, whether one policy can manage the band and the hedge jointly, and where active management pays.

Fees fail to cover impermanent loss at every width on the primary pool; the 0.30 % tier turns the wide bands net positive, while rebalancing loses on both. A standard learner collapsed into churn; after repairs, five seeds learned to deploy one band and hold. The mean learned policy fails the pre-registered Sharpe test against the frozen best static width ($-0.73$, CI $[-1.36, -0.04]$) and earns less, but cuts CVaR against every unhedged rule (+23.7 to +34.2). A delta-matched hedge cuts the tail to $-3.15$ from $-59.65$ for 2 731 per window. The best seed ties the static width on Sharpe and leads on tail risk. Active management buys insurance in stress and pays for it in calm.

**Index Terms:** reinforcement learning, Uniswap v3, concentrated liquidity, market making, loss-versus-rebalancing, impermanent loss, risk-sensitive reinforcement learning.

## I Introduction

Automated market makers let anyone earn trading fees without quoting prices by hand. Uniswap v3 sharpened the offer: a liquidity provider (LP) chooses the price range it is willing to trade in, so a deposit of 100 000 USDC earns the fee share of a larger full-range position while the price stays inside the chosen band.[1] Narrowing the band raises the fee earned per dollar of capital, and that promise drew liquidity into v3.

The promise hides a cost. Liquidity quoted away from the market price is an offer that arbitrageurs accept, and every such trade pushes the pool toward the market price while leaving the LP holding more

of the asset that is falling. Because the loss is a transfer rather than a fluctuation, it does not reverse when the price returns. Milionis et al. named this adverse selection cost loss-versus-rebalancing (LVR) and showed that a full-range position loses value at a rate proportional to the square of volatility.[2] Fritsch and Canidio measured realised arbitrage losses on large pools and found that they exceed the fees those pools earn,[3] and Loesch et al. showed that concentration magnifies the loss once the price leaves the band.[4] Fee income alone is not the LP's return.

An LP who wants more than a passive range has three decisions, and they interact. When to recenter trades fee income against gas and slippage; how wide to set the next band trades fee density against the time until an edge; how much of the position to hedge trades directional risk against funding and perp fees. Each decision is an impulse, costly and taken under uncertainty about volatility, order flow and regime. The objective is not to earn the most but to earn the most per unit of risk taken, which places the problem within risk-sensitive impulse control rather than static portfolio choice.[5]

Recent work takes up parts of it. Zhang et al. train a hedged agent for v3 provision, but the hedge executes without cost or funding and the policy sees no regime information.[6] ZeroSwap learns to set a market maker's spread so that quoted prices track an external reference, a different lever from range placement.[7] RAmmStein derives regime-dependent rebalancing thresholds and reports fewer rebalances for the same net return, but its net metric excludes impermanent loss and it works with off-chain data for a single pair.[5] Chionas et al. study risk aversion in concentrated liquidity inside a simulation and leave out on-chain data and the hedge.[8] None of these places timing, width and hedge decisions in a single risk-sensitive policy, charges every friction on both the liquidity and hedge legs, and evaluates it on recorded on-chain trades against hand-written rules.

This report fills that gap. We build a testbed on three years of recorded trades from two real ETH/USDC pools, with an exact replay backtester that computes fees in the protocol's own integer arithmetic, a fast simulator for training, and a parity harness that keeps the two honest on identical inputs. On that testbed we ask four questions.

- **Where do fees fail?** We decompose passive returns into fees, impermanent loss, gas and slippage, and locate the conditions under which the losses win.
- **Why does naive learning fail?** We document the collapse of a standard policy on this task, trace it to three interface problems, and measure what each repair buys.
- **Can one policy do better?** We train a risk-sensitive policy over timing, width and hedging and compare it out of sample with the best hand-written rules and a hedged benchmark.
- **Where is active management worth it?** We map the boundary between active and passive provision across volatility, fee tier, gas and capital.

The failures of the first policies are part of the result, not a footnote. In this domain, what the agent is allowed to see and what it is rewarded for matter more than the choice of learning algorithm, and a report that showed only the eventual policy would hide the more transferable lesson.

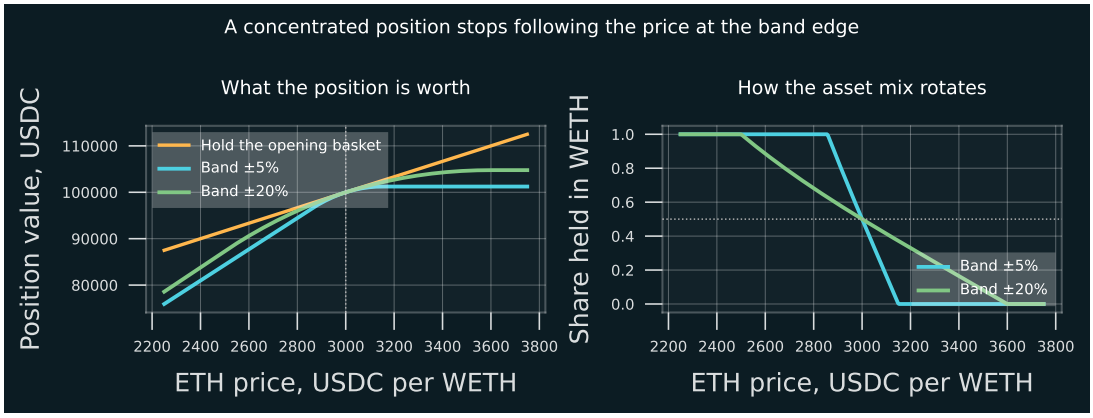



**Figure 1:** A concentrated position versus a passive holding of the same opening basket. Left: the position tracks the basket while the price sits inside the band and stops following it at the edges, so it gives up the upside of a rally while keeping the downside of a fall. Right: the asset mix rotates from WETH to USDC as the price rises and freezes at the edge. Illustrative parameters: 100 000 USDC entered at 3000 USDC per WETH.

# II How Concentrated Liquidity Works

## II.A A position as a bet on a range

Uniswap v3 indexes price by integer ticks. An LP picks a lower and an upper tick, and the deposit backs trades only while the pool price stays between them.[1] In protocol units the price is a raw square root, $s = \sqrt{1.0001^{\text{tick}}}$, which increases with the tick; the human price is $q(s) = 10^{d_1-d_0}/s^2$ USDC per WETH, so a lower tick means a higher ETH price. Inside a band $[s_a, s_b]$ the position holds the token amounts

$$x(s) = \frac{L}{10^{d_0}}\left(\frac{1}{s} - \frac{1}{s_b}\right), \tag{1}$$

$$y(s) = \frac{L}{10^{d_1}}\left(s - s_a\right), \tag{2}$$

and its USDC value is therefore

$$V(s) = x(s) + y(s)\,q(s) = \frac{L}{10^{d_0}}\left(\frac{2}{s} - \frac{1}{s_b} - \frac{s_a}{s^2}\right). \tag{3}$$

As the ETH price falls, the position sells USDC and accumulates WETH; as the price rises it does the opposite. At an edge of the band the position holds a single asset, and once the price leaves the band it stops trading altogether, so a passive position cannot escape the wrong side of a move.

Fees accrue in proportion to the position's share of active liquidity. Narrowing the band raises that share for a given deposit, which is why concentration is attractive, but it also shortens the expected time until the price reaches an edge. That tension between fee density and band lifetime sits underneath every decision in the report, and Fig. 1 shows the two effects together.

**Table 1:** The three levers and the trade-off each carries. Every return in the study is measured after the costs of the active levers are charged: the liquidity leg pays gas, slippage and impermanent loss, and the hedge leg adds funding and perp fees once the overlay is enabled.

| Lever | Choice | Buys | Costs |
|---|---|---|---|
| Timing $\kappa_t$ | hold or recenter | fee income on a band close to the price | gas and slippage; the fee position restarts |
| Width $w_t$ | width of the band, in tick spacings | fee density for a given deposit | time until the price reaches an edge |
| Hedge $h_t$ | short fraction of ETH delta | removal of directional swings | funding and perp trading fees |

## II.B Two losses, one larger

Impermanent loss (IL) is the shortfall of the position against holding the initial deposit. For a full-range position with equally valued legs and price ratio $r = q_t/q_0$,

$$\mathrm{IL}(r) = \frac{2\sqrt{r}}{1+r} - 1, \tag{4}$$

which is second order in the price move; at $r = 1.2$ it is $-0.41$ %. On v3 the same expression applies only inside the band, and outside it the position is a one-sided holding, so IL depends on where the band was placed.[4]

LVR is sharper. Compare the position with a portfolio that continuously rebalances to hold the same instantaneous exposure. A second-order expansion of the full-range value shows that the LP falls behind that portfolio at the rate

$$\mathrm{dLVR}_t = \frac{1}{8}\,\sigma^2\,V_t\,\mathrm{d}t, \tag{5}$$

the loss-versus-rebalancing of Milionis et al.[2] Adverse selection therefore grows with the square of volatility and with position size, and fee income must exceed it for passive provision to be viable. Concentration raises fee density but not the fee rate, so it helps only when the extra fee share outweighs the extra exposure to arbitrage; Section III finds that on many large pools it does not. Figure 2 shows the shape of both losses.

## II.C The decisions and the risk

An LP that wants to improve on the passive outcome must decide when to act, where and how wide to place the next band, and how much market risk to carry. The levers interfere, and Table 1 states what each one buys and what it costs. Because acting is lumpy, the problem is one of impulse control,[5] and because the outcome is uncertain the right measure is profit per unit of risk. We use the annualised Sharpe ratio and the conditional value-at-risk (CVaR) at the 5 % level as primary statistics; CVaR is coherent and has a convex representation that can be optimised directly.[9] Figure 3 makes the sequence of decisions and costs concrete.

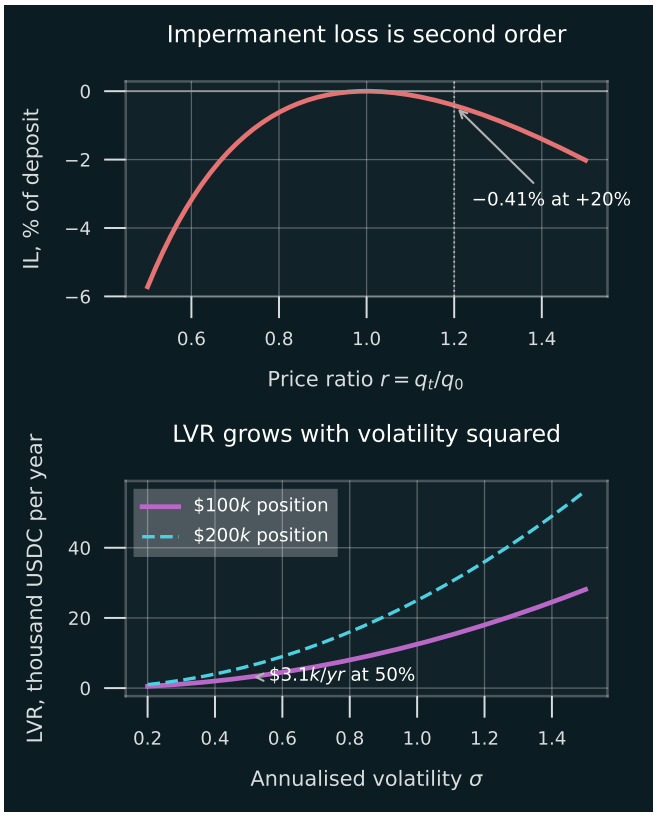



**Figure 2:** The two losses. Top: impermanent loss against the price move, second order and small for a full-range position. Bottom: the LVR rate against volatility for a 100 000 USDC and a 200 000 USDC position, showing both the quadratic dependence on volatility and the linear dependence on size.

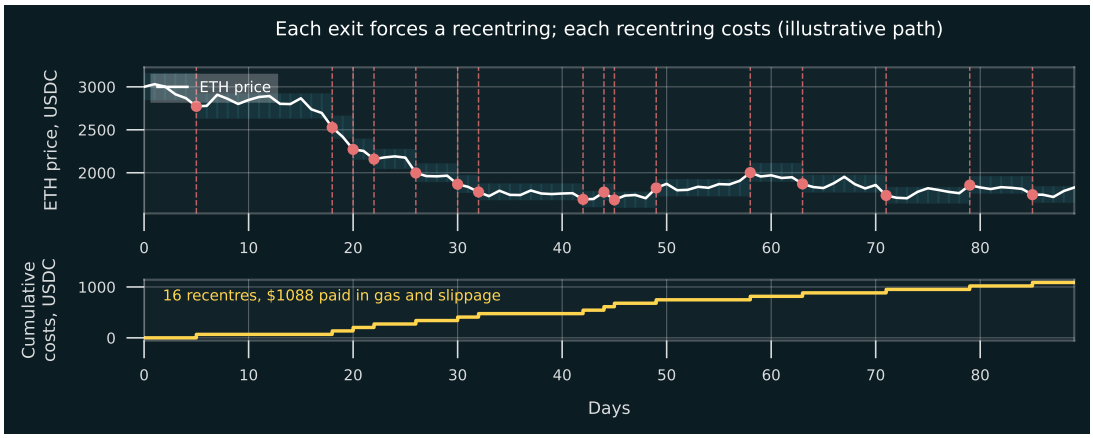



**Figure 3:** Why acting is a decision rather than a rule. An illustrative price path with a ±5% band: each exit forces a recentring, and each recentring adds gas and slippage to a cost that only ever grows. The path is illustrative and the costs are simplified; the study charges per-block gas and slippage on the inventory actually swapped.

# III Related Work

The economics of passive provision is now well understood. Milionis et al. formalised the loss a constant-product LP suffers against a delta-rebalanced portfolio and derived the rate $\frac{1}{8}\sigma^2 V$ for a full-range position; the loss appears on any price path, informed traders or not, and protocol fees and the choice of curve shift only the balance.[2] Fritsch and Canidio measured the losses on real pools and found that arbitrage losses typically dominate the fees earned, with passive full-range positions outperforming concentrated ones in their sample.[3] Loesch et al. studied v3 positions directly and showed that concentration magnifies impermanent loss when the price leaves the band.[4] Together these results define the terrain: gross fee yield is not an LP's return.

A second line of work learns the LP's decisions. Zhang et al. train a hedged double deep Q-network and report large gains over baselines, but the hedge is assumed frictionless and the policy sees no regime features.[6] ZeroSwap learns the spread of a market maker so that quoted prices track an external reference, an orthogonal lever to range placement.[7] RAmmStein poses regime-dependent rebalancing thresholds as a problem of impulse control and reports that rebalancing lazily when the regime warrants it improves net ROI with far fewer rebalances, though its net metric excludes impermanent loss and the study relies on off-chain data for one pair.[5] Chionas et al. treat concentrated liquidity as a risk-aversion problem in simulation and find that higher volatility and gas widen optimal positions and reduce rebalancing, while a risk penalty trims the adverse tail; their analysis uses no on-chain data and no hedge.[8] Classical market making supplies the risk-sensitive vocabulary we use, but its state is a limit order book and its actions are quotes, not range geometry.[10]

The learning machinery is standard. We use proximal policy optimisation with clipping (PPO),[11] implemented through Stable-Baselines3 on Gymnasium environments,[12,13] and evaluate risk with CVaR at the 5 % level through its Rockafellar–Uryasev representation.[9]

What is missing is one study that trains a single policy over the joint timing, width and hedge decisions, charges every friction on both legs, and evaluates it on validated on-chain replay under a risk-sensitive metric. That is the gap this report fills.

# IV The Liquidity Provider's Decision Problem

We model the LP's problem as a risk-sensitive Markov decision process with impulse control. Decisions are taken every ten minutes over an episode of thirty days. At each decision the agent sees a fixed observation, chooses between holding and acting, and receives a reward that measures how far the strategy's value has moved beyond a passive benchmark, net of every cost. The volatility regime and order flow are latent, so the process is partially observed: the agent sees only the history summarised in the observation. Figure 4 shows one decision cycle.

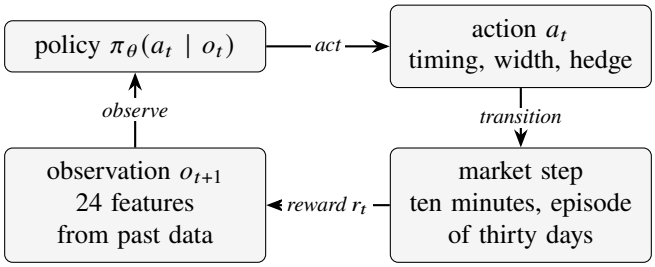



**Figure 4:** One cycle of the LP decision process. The policy acts every ten minutes; a decision to rebalance moves the band and charges gas and slippage, while a decision to hold leaves the position where it is. Rewards accumulate over an episode of thirty days against the fixed opening basket.

**Table 2:** The 24 observation features, grouped by source. Every field is computed from information available at the decision time, so an observation at time $t$ cannot contain a bar from $t + 1$.

| Group | Features | n |
|---|---|---|
| Market | price, sqrt price, tick, 1 h and 24 h log returns, realised 24 h volatility | 6 |
| Position | in-range flag, position value, two uncollected fee balances, IL against the fixed basket | 5 |
| Pool | active liquidity, fee tier | 2 |
| Costs | recent gas price, recent ETH reference price | 2 |
| Regime | regime label (bull, bear, sideways, high volatility) | 1 |
| Time | step in episode, steps remaining | 2 |
| Band | distance to each band edge and band width, in tick spacings | 3 |
| Hedge | short ratio, hedge equity, recent funding rate | 3 |
| | **Total** | **24** |

## IV.A State

The observation contains 24 features, all computed from information available at the decision time. Table 2 groups them by source. The band geometry deserves emphasis: the action space is written in tick spacings relative to the current tick, so a policy that cannot see how close the price is to an edge cannot time a rebalance at all. Heavy-tailed features such as price, liquidity and gas are compressed with a signed logarithm so that every dimension reaches the network on the same scale.

## IV.B Action

At each decision the agent selects a factored action

$$a_t = (\kappa_t,\, c_t,\, w_t,\, h_t), \tag{6}$$

whose four heads are described in Table 3. The study learns the width and hedge heads; the timing head is exercised once, at deployment, and the centre stays at the current tick. That restriction follows from the economics the first experiments measured: on this pool the cost of moving a band exceeds the fee a move can recover, so the timing decision collapses to the choice of whether to deploy at all. Section V reports the diagnostic runs that established this, and Section VII reports the learned policy under the restriction.

**Table 3:** The four action heads. Widths are given as half the band width, in tick spacings. The general grid holds ten widths; the study's configuration restricts the width head to three arms and the timing head to a single deployment.

| Head | Symbol | Options | Meaning |
|---|---|---|---|
| Rebalance | $\kappa_t$ | hold, act | spend gas and slippage to move the band, or leave it |
| Centre | $c_t$ | $-4, \ldots, 4$ spacings | shift the centre of the new band around the current tick |
| Width | $w_t$ | $\{1, 2, 5, 10, 25, 50,$ $100, 250, 500, 1000\}$ spacings | half of the new band's width |
| Hedge | $h_t$ | $\{0, 0.25, 0.5, 0.75, 1.0\}$ | share of the LP's ETH delta that the short offsets |

**Table 4:** Every term of the reward, in token0 (USDC) units before the $1/W_0$ scaling. Gas and slippage enter as positive magnitudes and carry a minus sign in the reward.

| Term | Meaning | Sign |
|---|---|---|
| $\text{fees}_t$ | fee income earned in the step | $+$ |
| $\text{gas}_t$ | gas for mints, burns, collects and rebalancing swaps | $-$ |
| $\text{slip}_t$ | slippage on the inventory actually swapped | $-$ |
| $\Delta(V_t - H_t)$ | change in position value against the fixed opening basket | $\pm$ |
| $\pi_t$ | mark PnL of the short perpetual | $\pm$ |
| $\phi_t$ | funding paid or received on the short | $\pm$ |
| $\tau_t$ | perp taker cost of adjusting the short | $\leq 0$ |
| $\lambda\, \hat{c}_t$ | penalty on the trailing expected shortfall of per-step losses, $\lambda = 0.05$ in the study | $-$ |

The width ladder used for the study holds three arms, one near the typical daily move, one just wider than the widest hand-written band, and one close to the full range. Together with the hold option they give the LP head four actions, and the hedge head adds the five ratios; a full action is therefore one of $4 \times 5$ pairs. The same interface admits the wider timing and centre grids, which remain available for the workstream on active timing.

## IV.C Reward

A rebalance converts inventory and pays for the privilege. Writing $V_t$ for the position value, $H_t$ for the fixed value of the opening deposit held passively, and the hedge cash flows as mark PnL $\pi_t$, funding $\phi_t$ and perp cost $\tau_t \leq 0$, the reward is

$$r_t = \frac{1}{W_0}\Big(\text{fees}_t - \text{gas}_t - \text{slip}_t + \Delta(V_t - H_t) + \pi_t + \phi_t + \tau_t - \lambda\, \hat{c}_t\Big), \tag{7}$$

whose terms are defined in Table 4. Dividing by the initial capital $W_0$ keeps value targets at a workable scale without hiding any cost. The excess is measured against the *fixed* opening basket: $H_t$ is never rebased at a rebalance, so the episode sum equals the excess over holding that basket, which is the quantity the evaluation reports.

### IV.D Risk and benchmarks

The learning objective is the discounted return with a penalty on the recent tail of per-step losses. The risk term in Eq. (7) subtracts $\lambda$ times the recent 5 % expected shortfall of per-step equity changes, measured over the last trading day, so deeper tail losses cost more reward. The penalty is computed on the joint equity of the liquidity and hedge legs, which is what lets the hedge ratio move it. The default configuration disables the term ($\lambda = 0$); the study's configuration sets $\lambda = 0.05$, small next to a full tail loss. The evaluation reports the same quantity the penalty controls. With losses $L = -G$ defined on the episode return $G$, the conditional value-at-risk at level $\alpha$ is

$$\mathrm{CVaR}_\alpha(L) = \min_{\nu \in \mathbb{R}} \left\{ \nu + \frac{1}{1-\alpha} \mathbb{E}\left[(L-\nu)^+\right] \right\}, \tag{8}$$

the Rockafellar–Uryasev representation,[9] which we report at $\alpha = 0.95$. Evaluation therefore reports the annualised Sharpe ratio and CVaR at 5 % as primary statistics, with raw PnL and excess over HODL as secondary ones. Comparisons use paired per-window differences and confidence intervals across seeds.

The hedge is a short perpetual position sized against the LP's ETH exposure. Its mark PnL over a step is the negative product of the short notional and the price change, plus funding and the taker cost of adjusting the position. Three reference points make the hedge leg interpretable: the unhedged LP, the policy that chooses the ratio, and a hedged HODL portfolio with a delta-matched short. The last is the benchmark that prevents us from declaring success against a weaker opponent: a policy that merely removes directional risk must still beat a passive portfolio with the same risk profile. The overlay is enabled in the study's configuration and its ratio is a learned decision at every step.

## V Building a Trustworthy Testbed

A study of this question needs a world in which the LP's costs can be measured rather than assumed. That world must be exact enough that a reported fee or loss is the real one, and fast enough that a policy can make millions of decisions. No single engine satisfies both, so the testbed uses two and checks them against each other. Figure 5 shows the flow.

### V.A The record

The data are the raw logs of two real ETH/USDC pools: the 0.05 % pool used as the primary market and the 0.30 % pool used as a fee-tier contrast. The pipeline turns those logs into one ordered event tape of swaps, mints, burns and collects, and joins it with three auxiliary feeds: one-minute reference prices, per-block gas prices, and regime labels computed from thresholds fixed in advance. A fourth feed, the funding rate of a perpetual margined in USDT, prices the hedge. The window runs from January 2022 to December 2024, with 2022–2023 for training and 2024 held out. The primary pool's training tape alone holds about 4.86 million events, which is why speed matters later. Every read goes

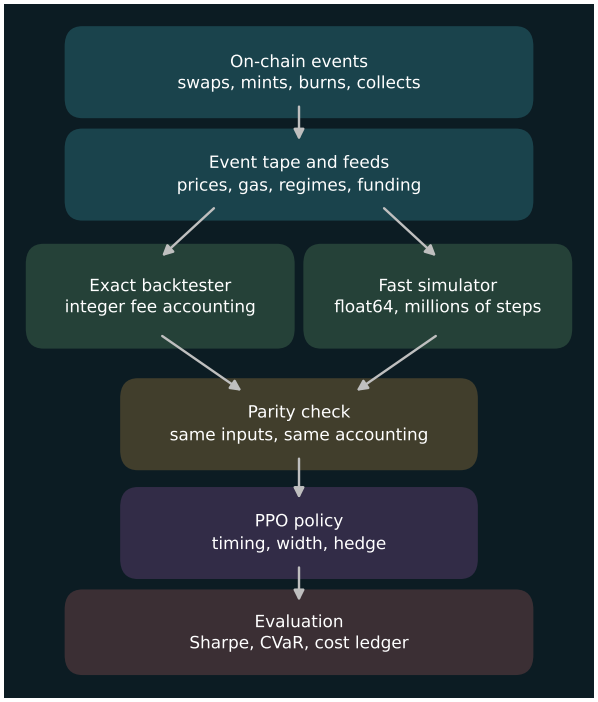



**Figure 5:** The testbed. Recorded events feed one tape; two engines consume it, an exact backtester for ground truth and a fast simulator for training; the parity check reconciles them before a policy is trusted.

through one interface that enforces block ordering and the train/eval boundary, so an observation at time $t$ cannot accidentally see a bar from $t + 1$.

## V.B The exact engine

The backtester replays a frozen policy against the tape in the protocol's own integer arithmetic. Fees come from the pool's recorded fee-growth accumulators, exactly as the contract computes them; liquidity events update an external tick lattice seeded from a boundary snapshot; swaps are apportioned across the ticks they cross. The ledger therefore satisfies an accounting identity by construction: fees plus impermanent loss, minus gas and slippage, equals the strategy's excess over holding the opening deposit. A backtested policy is frozen by construction because the interface exposes no way to update it, and every ledger records the configuration and seed that produced it.

## V.C Two engines, one check

Training on the exact engine would be far too slow, so the simulator mirrors the same economics in floating point. Its agent is marginal: the position never moves the price, and its fee share is the fraction of active liquidity it supplies. The simulator can run on the recorded events or on a calibrated regime-switching jump process fitted to the training years, and it executes each action at the price observed before the interval advances, as a live LP would.

Two numeric worlds are only useful if their disagreement is bounded. The parity harness feeds both engines the same prices, positions and gas series, and then compares fees, impermanent loss and PnL

**Table 5:** The defects that collapsed the first policies, and the repair for each. The first three are interface problems; the last two surfaced while fixing them.

| Defect | Consequence | Repair |
|---|---|---|
| Observation had no band geometry | the policy could not tell how close the price was to an edge or how far a recentring would move, so it could not time a rebalance | distances to each edge and the band width, in tick spacings |
| Reward measured loss against a rolling entry basket | after a rebalance the reward no longer matched the reported metric | excess against the fixed opening basket, never rebased |
| Every episode replayed one fixed window | the agent never saw a different market | training windows resampled from 2022–2023 |
| Slippage charged on the whole position value | costs overstated against a real swap | slippage on the inventory actually swapped |
| Fee credit taken from global fee growth | a position could be credited more than the fee actually taken when its liquidity dominated a thin segment | the position's exact share of each segment's LP fee |

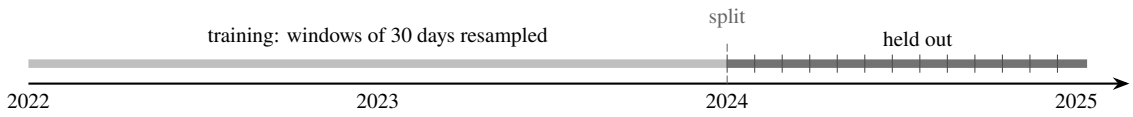



**Figure 6:** The walk-forward split. Training episodes are windows of thirty days sampled from 2022–2023; evaluation tiles 2024 into windows of thirty days, the last a partial period, and no episode crosses the split.

under a stated tolerance. Since the simulator's fee engine was made exact, the remaining drift on real data is tiny: mean fee differences on the order of $10^{-7}$ USDC per step per seed, and impermanent loss matching exactly. The harness originally exercised only the passive policy, which never rebalances; it now also runs a policy that does, which is the path the fee and loss results depend on. A separate suite of look-ahead probes attacks the composed system at the train/eval seam.

### V.D The failures that shaped the design

The first two training runs produced policies that did nothing useful: every seed collapsed to one deterministic action, some always holding and some always repeating a single rebalance, and the measured advantage barely distinguished the actions. The collapse was not a tuning problem. Table 5 lists the interface defects behind it and the repair for each. Each repair is small, but together they were the difference between a policy that can learn and one that cannot, which is why the failure is treated as a result rather than an implementation note.

## VI Experimental Design

The measurement protocol and the decision rule are fixed before the headline comparison; the interface corrections that preceded the final runs are reported as such. Table 6 states the pools, the split and the training budget, and Fig. 6 shows how the windows sit on the calendar.

**Table 6:** The study protocol and the training configuration. Both pools are USDC/WETH with USDC as token0; prices are quoted in USDC per WETH.

| Setting | Value |
|---|---|
| Primary pool | 0.05 % (500 pips), tick spacing 10 |
| Contrast pool | 0.30 % (3000 pips), tick spacing 60 |
| Training window | 2022-01-01 to 2023-12-31 UTC |
| Evaluation window | 2024-01-01 to 2024-12-31 UTC |
| Episode | thirty days, one decision every ten minutes |
| Capital | 100 000 USDC per episode |
| Evaluation tiling | 12 windows of 30 days plus a partial final window (13 episodes) |
| Seeds | 5 |
| Training budget | 1 000 000 timesteps per seed |
| Timing | one deployment per episode; the band is held afterwards |
| Width arms | half-widths of 50, 250 and 1000 tick spacings, plus hold |
| Hedge head | short ratio in $\{0, 0.25, 0.5, 0.75, 1.0\}$ at every decision |
| Risk penalty | CVaR at 5 % of the trailing one-day loss, $\lambda = 0.05$ |
| Checkpoint selection | highest validation Sharpe on the last 180 days of training |
| Algorithm | clipped PPO, $\gamma = 0.9999$, GAE $\lambda = 0.95$, clip 0.15 |
| Network | 2 hidden layers of 256 units |
| Parallel environments | 4 |

**Table 7:** Benchmark policies. The learned policy is compared with all of them; a positive claim requires beating the best static width and the delta-matched hedged benchmark, not just hodl.

| Benchmark | Rule |
|---|---|
| hodl | Hold the opening band of half-width 120 ticks; never recentre. |
| passive_narrow | Deploy a ±5 % band once; never recentre. |
| passive_wide | Deploy a ±20 % band once; never recentre. |
| full_range_v2 | The Uniswap-v2-equivalent full range. |
| tau_reset | Recenter whenever the price exits the live band. |
| cost_aware | Recenter at most daily, and only when the estimated fee gain exceeds gas. |
| Best static width | The ladder width with the best Sharpe on the training split, frozen for test. |
| Hedged HODL | The opening band held, plus a short perpetual matched to its delta, with funding and perp costs. |
| Learned policy | PPO over the joint (timing, width, hedge ratio) action space. |

## VI.A Benchmarks

A learned policy that beats only the weakest rule written by hand has demonstrated nothing. The comparison therefore includes the full ladder in Table 7, a best static width chosen on the training split and then frozen, and a hedged benchmark with a delta-matched short. Two reference holdings frame the ladder. A pure buy-and-hold basket holds the opening token mix as tokens and never provides liquidity; it is the numeraire for excess over HODL. The hodl rule instead deploys the opening band before the first decision and never recentres, so it earns fees and pays impermanent loss.

**Table 8:** The design choices the repair sequence touched, plus the targeted probes that complete the table. The repair changes were made in sequence rather than varied one at a time, so Section VII measures the sequence of repairs, not the marginal effect of each change; the friction, fee-tier and regime-feature rows are the probes.

| Design choice | Ablation | Question it answers |
|---|---|---|
| Band geometry | removed from the observation | does the agent need to see the live band to time rebalances? |
| Window sampling | windows held fixed instead of resampled | how much does resampling the training windows buy? |
| Hedge | overlay disabled | what does the hedge contribute net of funding? |
| Risk penalty | toggled | does the risk penalty change the tail without costing Sharpe? |
| Regime features | frozen to a constant across the five seeds | do the regime labels inform the policy? |
| Frictions | full model, no gas, no slippage, flat gas, no impermanent loss | which cost drives the result? |
| Fee tier | repeated on the 0.30 % pool | does a higher fee tier buy a longer viable range? |

## VI.B Metrics

Two primary statistics carry the inference, both computed on the joint equity curve of the liquidity and hedge legs. The annualised Sharpe ratio is the mean of the per-step equity returns over their standard deviation, scaled to a year at the ten-minute cadence. CVaR at the 5 % level is the average of the worst 5 % of per-step equity changes in USDC, the same quantity the reward penalty controls. A policy is compared with each benchmark on the per-window differences of both statistics, with bootstrap confidence intervals over the thirteen windows; because the tail is thin at that sample size, the worst window is reported alongside them. Secondary statistics are raw PnL, excess over HODL, annualised return, Sortino ratio, maximum drawdown, rebalance count, and the cost ledger of fee income, gas, slippage, realised impermanent loss and the hedge flows. Per-regime tables stratify by each window's dominant label: bull, bear, sideways or high volatility.

## VI.C Ablations and the boundary

The failure modes were repaired in sequence rather than varied in a factorial design. Table 8 lists the design choices the sequence touched: band geometry, the reward basket and window sampling, the rebalance interval, the hedge head and the risk penalty. Section VII reports the progression as a sequence of repairs, not as the marginal effect of each change. Three targeted probes complete the table: the friction terms are ablated on a one-shot band and on a recentring rule, the whole rule ladder is repeated on the 0.30 % tier, and an arm with the regime feature frozen is trained across the five seeds. For the boundary question the protocol calls for a sweep of realised volatility, volume relative to liquidity, fee tier, gas price and capital, evaluating each point on replayed windows rather than a fitted process. The completed sweep moves the gas and capital levels and the fee tier; the realised windows supply the volatility and volume variation the LVR overlay uses, while a controlled volatility and volume sweep remains open.

Decision rules are stated in advance: the learned policy earns a positive verdict only if its mean Sharpe beats the best static width and the hedged HODL with a paired 95 % confidence interval excluding zero, and its CVaR is no worse than the best heuristic's. Section VII reports where the rule is met and where it is not; where the evidence is negative, the negative result is the finding.

# VII Results

The four questions are answered in the order they were posed, on the 0.05 % (500 pips) ETH/USDC pool, over the thirteen windows of 2024 that no policy saw during training, and across five seeds where a learned policy is involved. Every number is measured on the exact backtester, with the frictions of the testbed charged on whichever leg is active. The tables name the window count and the units; where a mean is quoted in the prose, it is the mean over those thirteen windows.

## VII.A Where the fees fail

A passive position earns fees and pays for the price travel it cannot follow. Table 11 separates the two. The pattern is the same at every width: fees are positive and shrink as the band widens, impermanent loss is negative and shrinks faster, and the liquidity leg ends in deficit for every passive rule. The band held at the opening width nets −1553 USDC against the pure buy-and-hold basket; the narrow band nets −1206 USDC, the wide band −665 USDC, and the full range −106 USDC.

Concentration does exactly what the bargain advertises and no more. A half-width of 50 tick spacings collects 4266 USDC in fees against −5435 USDC of impermanent loss; the full range collects 362 USDC against −430 USDC. The ratio of fees to losses moves only from 0.78 to 0.84 across that span. Concentrating capital raises both terms together and leaves the shortfall in place; what changes is the variance of the outcome, not its sign.

The windows agree with the average. The opening band's liquidity leg is in deficit in six of the thirteen windows, the narrow band in five, the wide band in four and the full range in three. The losses cluster where the price trends or jumps. Across the five bull windows the opening band collects 6057 USDC of fees and pays −9481 USDC of impermanent loss; in the single high-volatility window fees of 1593 USDC meet a loss of −12 530 USDC. In the range-bound windows the sign flips and fees cover the shortfall with room to spare.

Acting does not repair the arithmetic; it deepens it. The two rebalancing rules buy more fee income than any one-shot rule by keeping the band close to the price, and they pay more for it in realised losses and execution costs. `tau_reset` recentres 122.8 times per episode, collects 43 029 USDC of fees, pays −48 904 USDC of impermanent loss and −14 093 USDC of gas and slippage, and ends −19 968 USDC behind the basket. The cost-aware rule trades 22.0 times per episode and still ends −4280 USDC behind it. `tau_reset` loses in every regime; the cost-aware rule earns only in the bull windows. On this pool in this year, fee income does not pay for the price risk a position carries, and frequent action adds cost without changing the sign. Whether a policy can exploit any of it is the next question.

**Table 9:** Friction ablation on the 0.05 % pool over the thirteen evaluation windows, mean PnL in USDC with Sharpe in parentheses. The one-shot rule is passive_wide; the recentring rule is tau_reset. no IL removes the impermanent-loss term, no gas and no slippage remove their cost term, and flat gas re-prices every rebalance at a flat 30 gwei base fee plus a 3 % tip at 3000 USDC per ETH.

| Variant | one-shot wide | recentring |
|---|---|---|
| full | +1 623 (1.53) | −17 680 (−5.81) |
| no IL | +5 172 (1.65) | +31 224 (12.84) |
| no gas | +1 659 (1.53) | −8 908 (−3.02) |
| no slippage | +1 624 (1.53) | −12 359 (−3.91) |
| flat gas | +1 583 (1.53) | −18 303 (−6.32) |

**Table 10:** The rule ladder on the 0.30 % (3000 pips) tier, mean over the thirteen evaluation windows. Net is the liquidity leg's excess over the opening basket; the two static rows are the price-equivalent of the primary pool's 250- and 1000-spacing arms at this pool's spacing of 60.

| Rule | PnL | Sharpe | Net |
|---|---|---|---|
| hodl opening band | +1 361 | 2.06 | −109 |
| passive_narrow (±5 %) | +1 432 | 2.89 | −38 |
| passive_wide (±20 %) | +1 808 | 3.33 | +338 |
| full_range_v2 | +2 188 | 1.30 | +719 |
| static width 42 spacings | +1 893 | 3.45 | +423 |
| static width 167 spacings | +2 204 | 1.97 | +734 |
| tau_reset | −20 702 | −15.89 | −22 172 |
| cost_aware | −2 101 | −0.27 | −3 571 |

The loss that does the damage is the one the position cannot avoid. Removing the impermanent-loss term from the ledger turns the churning rule from a 17 680 USDC loss into a 31 224 USDC gain and raises the one-shot band to 5172 USDC; removing gas and slippage moves the one-shot band by less than 40 USDC but the churner by 14 093 USDC. The one-shot rule loses to the basket because of price travel, not because of execution, and the recentring rule adds execution to a loss it has already made larger.

The fee tier changes the sign. On the 0.30 % pool the wide, full-range and static-width arms net between 338 USDC and 734 USDC against the opening basket, because a larger fee per unit of volume finally covers the loss to informed flow; the two rebalancing rules lose more than 2000 USDC there as well. The contrast shows what the primary pool would need: the same widths, on a tier that pays six times as much per swap, cross from deficit to surplus.

## VII.B Why the first policies failed

The first question asked of the learner was whether it could act at all, and on the original interface it could not. With the timing, centre and width decisions encoded as 91 actions and the reward measured against a basket that moved with each rebalance, every trained policy churned: it shifted the band every few steps, paid gas and slippage on each move, and lost −86 889 USDC across the evaluation windows. Shrinking the grid to 28 actions changed the magnitude, not the behaviour; that pilot lost −80 642 USDC. The collapse was not an optimisation accident. The reward gradient was dominated

**Table 11:** Fee income against impermanent loss on the thirteen evaluation windows, mean per window in USDC. Fees, impermanent loss, gas and slippage are measured against the pure buy-and-hold basket, so the liquidity net excludes the basket's own 2288 USDC and the hedge column is zero for the unhedged rules. The last column counts windows in which the liquidity net is negative.

| Policy | Fees | IL | Costs | LP net | Hedge | Net<0 |
|---|---|---|---|---|---|---|
| hodl | +4 734 | −6 287 | 0 | −1 553 | 0 | 6/13 |
| passive_narrow | +4 266 | −5 435 | −37 | −1 206 | 0 | 5/13 |
| passive_wide | +2 921 | −3 549 | −37 | −665 | 0 | 4/13 |
| full_range_v2 | +362 | −430 | −37 | −106 | 0 | 3/13 |
| tau_reset | +43 029 | −48 904 | −14 093 | −19 968 | 0 | 13/13 |
| cost_aware | +19 833 | −22 038 | −2 075 | −4 280 | 0 | 10/13 |

by the decision to act or hold, and the observation gave the policy no way to see where the price sat inside its band.

The repairs were applied in sequence rather than as a factorial ablation, so the progression below measures the sequence of repairs, not the marginal effect of any single change. Three interface defects were fixed first: the observation gained the distance to each band edge and the band width; the reward was measured against the fixed opening basket instead of a basket that moved at each rebalance; and training windows were resampled from 2022–2023 instead of replaying one fixed month. Two further defects surfaced while making those repairs. Slippage was charged on the whole position value rather than the inventory actually swapped, and fee credit was taken from global fee growth rather than the position's exact share of each liquidity segment. Both were corrected before any result reported here.

Three changes to the learning setup followed. Stabilising the update targets made every seed hold for the whole episode with zero rebalances and a mean episode return between −0.17 and −0.29, against −0.95 to −0.97 in the churning runs; the update stayed inside its trust region, with the KL divergence between 0.001 and 0.002 and the clip fraction near 0.02. The policy had learned to do nothing, which tied hodl at 735.2 USDC. Reducing the action space to a one-shot band choice and lengthening the effective horizon gave the width decision a signal at last. The first pilot that genuinely chose a band scored 2146.6 USDC with a Sharpe of 0.942 and a maximum drawdown of 9.7 %, matching the full-range rule's PnL with a higher Sharpe ratio and beating hodl by roughly a factor of three.

The hedge head failed in its first joint run. Every seed held the liquidity position for the whole episode and collapsed the hedge ratio to a constant that differed by seed, between 0.25 and 0.75. The seed selected on the training curve, which had settled at 0.75, lost 1313 USDC with a Sharpe of −1.294 out of sample. The objective had never asked for tail protection, and selection had used the training return rather than a validation statistic. Adding a penalty on the conditional tail of joint equity and selecting checkpoints by validation Sharpe produced the final policy whose results fill the rest of this section.

The regime label is not what the policy learns from, but it shapes what it learns. A second arm trained across the five seeds with the regime feature frozen to a constant reaches a mean Sharpe of 0.837 and a mean CVaR of −32.89 USDC, against 0.563 and −27.63 USDC with the live label. The Sharpe difference is not distinguishable at this sample size (paired interval $[-0.13, +0.68]$) while the ablated arm's CVaR is 5.3 USDC worse (paired interval for the difference $[-6.2, -4.2]$), so the live regime

feature buys tail protection rather than return. The ablated seeds also diverge more, from one that never deploys to two that widen to the 1000-spacing band, and the arm's best validation seed finishes worst out of sample: the selection signal is thin even when the policy learns.

### VII.C Whether one policy does better

The pre-registered comparison runs against the strongest hand-written rules, the best static width chosen on the training split, and a hedged benchmark with a matched delta. Table 12 sets out the levels and Table 13 the paired per-window differences that the decision rule turns on.

Start with the frozen comparator. Training Sharpes for half-widths of 50, 250 and 1000 tick spacings were 2.20, 1.45 and 0.80, so the protocol froze the narrowest band. On the evaluation windows that choice scores a Sharpe of 1.296, while the width of 250 would have scored 1.591. The frozen width is the pre-registered comparator, and the learned mean loses to it: the paired Sharpe difference is $-0.733$ with an interval of $[-1.355, -0.041]$ that excludes zero. The mean also loses to the wide band by 0.963, again with an interval excluding zero. The Sharpe clause of the pre-registered rule fails.

The tail clause holds against every unhedged benchmark. The learned mean's CVaR is $-27.63$ USDC. hodl sits at $-59.65$ USDC, the frozen width at $-56.95$ USDC, the wide band at $-51.31$ USDC and the full range at $-53.18$ USDC. Every paired CVaR difference against an unhedged benchmark in Table 13 is positive with an interval that excludes zero. Against the delta-matched hedge the clause fails: the hedge reaches a CVaR of $-3.15$ USDC, and the learned mean is $-24.48$ USDC worse, with a paired interval from $-30.42$ to $-18.95$. Both clauses of the rule must hold for a positive verdict, so the study reports none. The evidence supports a reduction in tail risk, not a superior return.

The mean hides the best seed. The seed with the highest validation Sharpe during training, seed three, deployed a half-width of 250 tick spacings for every window and held it. It returns 175.13 USDC, a Sharpe of 1.425 and a CVaR of $-17.04$ USDC. Its Sharpe is indistinguishable from the frozen width (difference +0.129, interval $[-0.880, +1.232]$) and its CVaR is better by 39.91 USDC with an interval of $[+27.76, +51.43]$. The same seed beats the delta-matched hedge on Sharpe by 6.19 with an interval that excludes zero, and loses to it on CVaR by $-13.89$ USDC. A single seed is not the pre-registered unit, and a seed chosen after seeing the evaluation windows could not carry the verdict. That seed did have the best validation Sharpe of the five on the training split, so at least its selection did not use the evaluation year.

The seeds agree on the timing decision and disagree on width and hedge. Every seed deploys once, at the start of the episode, and holds. The action distribution in Table 14 records one rebalance per episode for all five seeds: three choose the half-width of 250 tick spacings in every window, one splits between 250 and 50 across the tiling, and one chooses 50 throughout. The hedge ratios are partial and differ by seed, mostly between 0.25 and 0.75. Seed three holds 0.75 for 73.8 % of its steps, seed two splits between 0.25 and 0.75, and seed zero holds 0.50 or 1.00. The learned policy sets a band and a hedge ratio; it is not a timing policy, and its preferred width of 250 is the static band with the best Sharpe on the evaluation windows, though it was not the frozen comparator.

**Table 12:** Performance in the 2024 evaluation on the 0.05 % (500 pips) ETH/USDC pool, mean over the thirteen windows of the 2024 tiling. PnL and excess are USDC per episode; excess is measured against the pure buy-and-hold basket, the study's HODL numeraire. The row for the learned mean averages the five seeds; the row for the best seed reports the seed with the highest validation Sharpe on the training split. Rebalances counts executed moves per episode. The frozen comparator selected on the training split is the half-width of 50 tick spacings, the passive_narrow row.

| Policy | PnL | Excess | Sharpe | CVaR | maxDD | Reb. |
|---|---|---|---|---|---|---|
| Basket | +2 288 | 0 | 0.441 | −53.81 | −9.6 % | 0 |
| hodl | +735 | −1 553 | 0.654 | −59.65 | −12.2 % | 0 |
| passive_narrow | +1 082 | −1 206 | 1.296 | −56.95 | −11.3 % | 1 |
| passive_wide | +1 623 | −665 | 1.526 | −51.31 | −10.1 % | 1 |
| full_range_v2 | +2 183 | −106 | 0.639 | −53.18 | −9.7 % | 1 |
| tau_reset | −17 680 | −19 968 | −5.810 | −56.10 | −21.9 % | 122.8 |
| cost_aware | −1 992 | −4 280 | −0.241 | −61.84 | −14.0 % | 22.0 |
| Hedged hodl | −1 996 | −4 284 | −4.766 | −3.15 | −2.7 % | 0 |
| Learned mean | −310 | −2 599 | 0.563 | −27.63 | −6.0 % | 1 |
| Best seed | +175 | −2 113 | 1.425 | −17.04 | −3.2 % | 1 |

**Table 13:** Paired per-window differences, learned mean minus benchmark, with bootstrap 95 % intervals over the thirteen windows; a star marks an interval that excludes zero. A positive Sharpe difference favours the learned policy; a positive CVaR difference means a smaller tail loss than the benchmark. Units are USDC for the CVaR column.

| Benchmark | Sharpe | CVaR |
|---|---|---|
| hodl | −0.091 [−0.808, +0.750] | +32.019 [+21.319, +42.230]* |
| passive_narrow | −0.733 [−1.355, −0.041]* | +29.320 [+19.365, +38.765]* |
| passive_wide | −0.963 [−1.472, −0.509]* | +23.677 [+18.298, +29.475]* |
| full_range_v2 | −0.076 [−0.930, +0.693] | +25.549 [+17.841, +33.977]* |
| tau_reset | +6.373 [+4.702, +8.278]* | +28.471 [+21.456, +36.160]* |
| cost_aware | +0.804 [−0.222, +1.656] | +34.207 [+26.501, +42.092]* |
| Hedged hodl | +5.329 [+2.550, +7.437]* | −24.481 [−30.423, −18.945]* |

The worst window for the one-shot rules and the learned mean is the single high-volatility window. hodl loses 22 999.6 USDC there and the frozen width 22 542.2 USDC. The wide band loses 19 123.1 USDC and the full range 12 641.5 USDC. The learned mean loses 10 039 USDC with a CVaR of −46.5 USDC. Raw PnL tells the same story as the tail across the year: the best unhedged benchmark, the full range, finishes 105.6 USDC below the pure basket, and the learned mean finishes 2599 USDC below it. No paired PnL difference against the ladder excludes zero except the comparison with tau_reset, so the return evidence is inconclusive at this sample size.

## VII.D Where active management pays

The regime map is the last answer, and it maps insurance rather than outperformance. Table 16 splits the evaluation windows by their dominant label. The counts are uneven by construction, with five bear and five bull windows, two sideways windows and a single high-volatility window, so the high-volatility row of the table is a case study rather than a distribution.

The quiet windows pay for feet that stay still. In the two sideways windows hodl earns 4648 USDC

**Table 14:** Deployment of each seed on the thirteen evaluation windows. Bands are half-widths in tick spacings, with the number of windows in which the seed chose that band; hedge ratios are shares of decision steps at each short fraction. Every seed rebalanced exactly once per episode.

| Seed | Band (windows) | Hedge ratio (share of steps) |
|---|---|---|
| 0 | 50 (13) | 0.50: 69.5 %, 1.00: 30.5 % |
| 1 | 250 (12), 50 (1) | 0.50: 51.0 %, 0.25: 27.9 %, 0.75: 14.6 %, 1.00: 6.5 % |
| 2 | 250 (13) | 0.25: 58.2 %, 0.75: 41.8 % |
| 3 | 250 (13) | 0.75: 73.8 %, 0.50: 22.1 %, 1.00: 3.4 %, others: 0.7 % |
| 4 | 250 (13) | 0.25: 59.5 %, 0.75: 18.8 %, 1.00: 12.9 %, 0.00: 8.7 % |

with a Sharpe of 1.73, the narrow band 3558 USDC with 1.79, and the delta-matched hedge subtracts 3145 USDC; the learned seed finishes at 74 USDC with a Sharpe of 0.41. Fee income is at its most reliable when the price does not travel, and in those windows every unit of hedge costs more than it protects.

The trending windows split by direction. In the five bull windows the full range earns 7379 USDC and hodl 4335 USDC, while the learned seed earns 1093 USDC with a Sharpe of 3.13: the hedge gives up part of the rally, and that is what a short does. In the five bear windows the ordering changes. The wide band earns 2589 USDC by holding USDC through the fall and the frozen width 1772 USDC, while the learned seed earns 527 USDC with the best risk-adjusted profile on the panel, a Sharpe of 1.09 against $-0.51$ for hodl and $-0.53$ for the full range. The learned seed's bear CVaR is $-19.6$ USDC against $-78.3$ USDC for hodl.

The single high-volatility window is where active management earns its premium. Every unhedged rule loses between 12 642 USDC and 22 999.6 USDC; the learned seed loses 5969 USDC, about a quarter of the hodl loss, and its CVaR is $-27.8$ USDC against $-94.5$ USDC. No policy's Sharpe is positive in that window, including the learned one at $-3.40$. The delta-matched hedge loses 1330 USDC with a CVaR of $-1.4$ USDC, the smallest loss and the largest hedge cost over the year. Widening the band or hedging the delta makes the stressed quarter hurt less; neither choice pays for itself in the quiet ones.

The controlled sweep moves along both sides of the boundary. Table 15 reruns the ladder at a quarter and four times the recorded gas price and at a tenth and ten times the capital. No level turns the recentring rules positive in PnL. tau_reset runs from $-11\,613$ USDC at quarter gas to $-40\,032$ USDC at four times the recorded price, and cost_aware, whose gate suppresses moves as gas rises, stays between $-2953$ USDC and $-1419$ USDC across the gas levels. Cheap execution does not revive timing, and a larger book does not either, because fees and impermanent loss scale with it while the fixed gas advantage stays small. The learned seed deploys once at every level and holds; its Sharpe ratio stays between 1.40 and 1.46.

The fee tier is the axis that moves the passive sign, and the analytical LVR rate says why. With $0.125\,\sigma^2 Vt$ as the floor under the value a liquidity position loses to arbitrage,[2] the opening band's fee income covers that floor in every full window, and in the partial final window once the floor is

**Table 15:** The boundary sweep on the 0.05 % pool, mean over the thirteen evaluation windows: PnL in USDC with Sharpe in parentheses. The gas levels scale the per-action gas cost; the capital levels scale the book. The last column is the best validation seed of the main run.

| **Level** | hodl | wide | tau_reset | cost_aware | best seed |
|---|---|---|---|---|---|
| Gas ×0.25 | +735 (0.65) | +1 651 (1.53) | −11 613 (−3.72) | −2 953 (−0.54) | +203 (1.42) |
| Gas ×1 | +735 (0.65) | +1 623 (1.53) | −17 680 (−5.81) | −1 992 (−0.24) | +175 (1.42) |
| Gas ×4 | +735 (0.65) | +1 509 (1.53) | −40 032 (−12.85) | −1 419 (0.01) | +65 (1.42) |
| Capital 10k | +81 (0.68) | +128 (1.53) | −6 764 (−15.37) | −195 (−0.10) | −13 (1.46) |
| Capital 100k | +735 (0.65) | +1 623 (1.53) | −17 680 (−5.81) | −1 992 (−0.24) | +175 (1.42) |
| Capital 1M | +1 384 (0.46) | +16 213 (1.51) | −162 687 (−5.47) | −54 962 (−1.35) | +2 003 (1.40) |

**Table 16:** Mean PnL (top panel) and mean CVaR at 5 % (bottom panel) in USDC by dominant regime label over the 2024 evaluation windows: five bear, five bull, two sideways and one high-volatility window. The Width 50 and Width 202 columns are the narrow (±5 %) and wide (±20 %) one-shot bands, whose mean half-widths are 50 and 202 tick spacings. The column for the best seed reports the seed with the highest validation Sharpe on the training split. The single high-volatility window is a case study, not a distribution.

| | **Mean PnL (USDC)** | | | | | |
|---|---|---|---|---|---|---|
| **Regime (windows)** | hodl | Width 50 | Width 202 | Full range | Delta hedge | Best seed |
| Bear (5) | +317 | +1 772 | +2 589 | +1 044 | −1 707 | +527 |
| Bull (5) | +4 335 | +4 127 | +4 779 | +7 379 | −1 958 | +1 093 |
| Sideways (2) | +4 648 | +3 558 | +1 693 | −550 | −3 145 | +74 |
| High volatility (1) | −23 000 | −22 542 | −19 123 | −12 642 | −1 330 | −5 969 |
| | **Mean CVaR at 5 % (USDC)** | | | | | |
| Bear (5) | −78.3 | −74.3 | −57.3 | −49.2 | −2.0 | −19.6 |
| Bull (5) | −31.9 | −31.2 | −37.0 | −58.3 | −4.1 | −12.8 |
| Sideways (2) | −65.0 | −59.3 | −50.8 | −49.6 | −4.4 | −15.8 |
| High volatility (1) | −94.5 | −94.4 | −93.9 | −54.5 | −1.4 | −27.8 |

scaled to its shorter length; the thinnest full-window margins (ratios of 1.5 and 1.8) fall in the two windows with the least fee income. On this fee tier the same widths still lose to the opening basket, and the 0.30 % tier is where they cross into surplus. The evaluation year holds one high-volatility window, so the stress row stays a case study and the boundary drawn here holds for this pair, this year and the tested ranges.

# VIII Discussion

## VIII.A The boundary the windows draw

The four answers converge on a single boundary, and stress draws it. Passive provision paid in the quiet windows: in the sideways windows the hodl earned 4648 USDC, the narrow band kept most of it, and the delta-matched hedge subtracted 3145 USDC. Breadth mattered more than action in the trending windows, because a wide band carries the market exposure that a narrow band sells away, and the full range earned 7379 USDC in the bull windows against 1093 USDC for the learned seed. The one high-volatility window reversed the ordering: every unhedged rule lost between 12 642 USDC and 22 999.6 USDC, the learned seed lost about a quarter of the hodl loss, and the full hedge lost the least.

The learned policy sits on the defensive side of that boundary, and the premium it pays is visible in every calm window. Judged on tail risk the boundary moves in its favour; judged on PnL it stays with the passive bands.

The regime labels carry only the windows they label, and one high-volatility window cannot estimate a regime. The map is a description of the evaluation year, not a fitted boundary, and the study's inference rests on the paired intervals of the primary statistics rather than on any single regime cell.

### VIII.B What the hedge buys and costs

The hedge is insurance, and the study compares policies that sit at three points along a single trade-off between protection and cost, though not as a controlled hedge ablation. At full delta match on the opening band the floor is nearly flat: CVaR falls from −59.65 USDC to −3.15 USDC and the maximum drawdown from 12.2 % to 2.7 %, and the bill is an average −2731 USDC of hedge flows and a total PnL of −1996 USDC. At the learned seed's partial ratios the middle of that trade-off appears: CVaR −17.04 USDC and a hedge cost of −1557 USDC. The one-shot unhedged rules sit at the other end, with CVaR between −51.31 USDC and −59.65 USDC and no premium; the two rebalancing rules trade at −56.10 USDC and −61.84 USDC. Across the compared policies the ordering is monotone in protection and in cost. Funding was positive on average at every point because the short collected it; the mark loss on a short in a rising market and the perp fees outweighed that income.

The hedge does not improve the mean, it reshapes the tail, so its value depends on the risk budget the LP is managed against. A mandate that treats tail risk as important as Sharpe will buy the protection that a mandate maximising return will refuse. The pre-registered rule gives tail risk equal standing, which is why a policy that fails the return clause and passes the tail clause is a partial success rather than a failed idea.

### VIII.C Why timing collapsed to a deployment decision

The learned policy never learned to time; its seeds deployed one band and held it. The rebalancing benchmarks show why. On this fee tier and in this year, frequent recentring realises impermanent loss again and again, and the fees collected do not cover it: `tau_reset` recentred 122.8 times per episode and ended 19 968 USDC behind the basket, the cost-aware rule recentred 22.0 times and ended 4280 USDC behind, and the one-shot rules ended between 106 USDC and 1553 USDC behind. Timing is not irrelevant in principle; it is dominated in this market.

The study's configuration restricted the timing head after the early runs showed the collapse, so the estimator was never asked to learn a decision that the evidence says it could not afford. That restriction is itself a finding about the pool, and it bounds the claim: a higher fee tier, cheaper execution or a market where price moves arrive more slowly could move the boundary, and the width choice is the substitute for timing under those conditions. The action interface still admits the wider timing grid, so a future run can test the shift without a redesign.

**Table 17:** Threats to validity and the scope each imposes.

| Threat | Scope |
|---|---|
| One asset pair at two fee tiers | the boundary is measured within that scope and does not automatically transfer to other pairs |
| CVaR at 5 % from few episodes | reported with bootstrap intervals and the worst window, not treated as decisive alone |
| Regime labels from fixed thresholds | a labelling device for reporting and training, not a discovered state |
| Boundary sweep over several axes | inference rests on paired intervals on the pre-registered primary metric; secondary wins are labelled as such |
| Simulator and backtester both replay recorded states | claims hold under the stated cost model, not for live execution |

The study's evidence has a shape that limits it. The failure modes were repaired in sequence rather than varied one at a time, so the ablation table reports the sequence and three targeted probes, not the marginal effect of every design choice. The boundary is drawn from the realised 2024 windows on one pool pair; the 0.30 % tier is measured for the rule-based ladder but not retrained with the learned policy; and the regime with a single window is a case study. What remains open is a factorial ablation of the repairs, a controlled volatility and volume sweep, and a learned policy on the contrast tier, all re-runs of the same testbed.

### VIII.D Threats to validity

The simulator replays recorded events, so prices, liquidity and fee growth are real rather than modelled. Its assumption that the agent is marginal is the main simplification: the position never moves the price, and its fee share is proportional to active liquidity. That is the standard replay assumption and it is recorded with every run, but it cannot capture queueing at the tick level or the impact of an unusually large position. The gas model charges representative per-action units rather than transaction receipts, so gas is a defensible floor rather than an exact quote, and the hedge uses a perpetual proxy whose funding and basis are representative rather than exact for a specific venue.

Table 17 states the threats that bound what the study can claim and the scope each imposes. The parity check bounds the distance between the simulator and the backtester, not the distance between the backtester and reality. Where the data allow, the external part of the gap is quantified by reconciling fees and collects; where they do not, latency, the privacy of order flow and MEV are discussed rather than measured. We therefore claim results about performance under risk on recorded market states under a stated cost model. We do not claim deployable profitability, optimality of the action grid, or that the policy would survive market conditions absent from 2022 to 2024.

## IX Conclusion

Concentrated liquidity promises more fee income while the price stays inside a chosen band, and the same position that earns those fees is the position that loses to informed order flow. An LP who wants

to do better than accept that must decide when to recenter, how wide to stay and how much risk to hedge, and each decision is expensive and uncertain.

This report measures whether a risk-sensitive learned policy can make those decisions jointly better than the best rules written by hand, and where active management stops paying. Answering that required a world in which costs are measured rather than assumed, so we built a testbed from three years of recorded trades, with an exact replay engine for ground truth, a fast simulator for training, and a parity harness between them. The study then asked four questions: where fees fail, why a standard agent fails, whether a learned policy wins out of sample, and where the boundary between active and passive provision lies.

The measured verdict is narrower than the bargain promised. On the 0.05 % pool in 2024 the liquidity leg of every rule ran a deficit against the pure buy-and-hold basket, and no strategy in the ladder beat that basket on PnL; the best unhedged rule, the full range, finished 105.6 USDC below it. Active rebalancing fared worst: `tau_reset` lost 17 680 USDC and the cost-aware rule 1992 USDC, while deploying one frozen static band earned 1082 USDC. The learned policy reproduced the width its seeds preferred without learning to time, failed the pre-registered Sharpe test against that frozen width (difference $-0.733$, interval $[-1.355, -0.041]$), and passed the tail test against every unhedged rule. Its best seed tied the static width on Sharpe and led it on CVaR, and the delta-matched hedge bought a flat tail at a carry cost. On this market, learning did not make active management pay; it produced a policy whose value is defence, and it priced the insurance.

The boundary of the study is also a map for future work: the action space can be extended with continuous width and an explicit exit decision, the boundary sweep can cover more pairs and fee tiers, and latency, private order flow and the exact funding of a real venue can be modelled rather than proxied. Each extension is a re-run of the same testbed rather than a redesign of it, which is the point of building the testbed before asking the question.

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